MIL-E-36359(EM)
14 January 1965

MILITARY SPECIFICATION

ELECTROENCEPHALOGRAPH, 8 CHANNEL, 110 VOLT, 60 CYCLE, AC

1. SCOPE

1.1 This specification covers one type of mobile electroencephalograph suitable for clinical encephalography in military medical treatment facilities.

2. APPLICABLE DOCUMENTS

2.1 The following document of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein:

SPECIFICATION

Federal

W-C-596 Connector, Plug, Electrical; Connector, Receptacle, Electrical.

(Copies of specification required by suppliers in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Report of the Committee on Apparatus of the International Federation of Societies for Encephalography and Clinical Neurophysiology as reported in the Electroencephalography Journal, Amsterdam, Netherlands, Volume 10, No. 2 May 1958.

FSC 6515

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(Copies of the report required by suppliers in connection with specific procurement functions should be obtained from the procuring agency).

3. REQUIREMENTS

3.1 Fire and casualty hazards. The electroencephalograph shall comply with the fire and casualty hazards provisions of the applicable standards and be an approved type for the purpose intended, as determined by a nationally recognized testing agency, adequately equipped and competent to perform such services, as shall be evidenced by the attachment of its seal, stamp or label to the article, or issuance of a certificate. The stamp, label, seal or certificate of the Underwriters' Laboratories (New York, N.Y.), the United States Testing Laboratories, Inc. (Hoboken, New Jersey), the Electrical Testing Laboratories, Inc. (New York, N.Y.), or other testing laboratory approved by the contracting agency shall be considered acceptable evidence of such approval; however, in addition to the aforementioned, the contracting agency reserves the right to require the contractor to submit a laboratory test report which shall include test results and methods used to determine conformance.

3.1.1 Compliance with the above requirements regarding fire and casualty hazards, does not absolve the contractor from complete compliance with the requirements of this specification in order to secure the acceptance of his commodity.

3.2 Material. Material used in the construction of the electro-encephalograph shall be structurally and electrically satisfactory to maintain operational alignment for the life of the equipment.

3.3 Design. Electroencephalograph (EEG) shall be an eight channel mobile type unit equipped with casters suitable for ready movement and shall operate from a nominal 117 volt, 60 cycle, AC power source and be insensitive to plus or minus 10 percent variation in line voltage. The EEG shall meet the Electrical Mechanical Specifications and the construction and General Requirements for Electroencephalographs included in the Report of the Committee on Apparatus of the International Federation of Societies for Encephalography and Clinical Neuriphysiology as reported in the Journal EEG, Clin Neurophysiol Volume 10, No. 2, pages 378 - 380, May 1958. All electrical components shall be hermetically sealed, and every cable and receptacle shall bear a matching label. The unit shall be a hybrid type, or as an alternate, may be fully transistorized, and shall contain plug-in modular type components. A battery for calibration purposes, time marker and master gain control shall also be included.

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3.4 Construction. The electroencephalograph shall contain a chart drive, writer system, electrode selector panel, electrode board, master switch, and a 23 point pushbutton operated input selector. The amplifiers, gear box, choppers (if used), electrode selector panel, master switch, power supply heat sinks, calibration battery, oscillographs. and regulated power supply shall be plug-in units.

3.4.1 Amplifiers. Amplifiers shall be hybrid, and provide an input for DC signals of 1 volt, full scale. The input rejection ratio shall be not less than 5,000 to 1. A sharp 60 cycle elimination filter shall be provided. It shall be possible to measure tube and transistor voltages at pin jacks on the amplifier panels. All power, input and output connections shall be automatically made when the amplifier is plugged in. Output jacks along the amplifier panels shall provide signals for testing the tube and transistor voltages at the various stages of amplification.

3.4.2 Electrode selector panel. Electrode selector panel (Numeric system) shall have at least 23 active positions including two reference jacks labeled A1 and A2 and a ground jack labeled GND. Electrode selector panel (International system) shall in addition to including positions specified for the Numeric system contain two nasopharyngeal leads. The panel shall contain provisions for connection of a volt-ohmmeter for determination of voltage and electrode resistance.

3.4.3 Master switch. Master preselector switch shall be a plug-in type for automatically programming runs between various electrode combinations. Upon notice of award, the successful bidder shall forward to each receiving activity a blank master switch programming chart for completion by the responsible medical officer. The completed chart shall indicate the program of specific electrode combinations desired by the activity for routine EEG surveys.

3.4.4 Calibration. Calibration voltages shall be derived from a long life plug-in mercury cell battery. Calibration voltages shall be 5, 10, 20, 50, 100, 200 and 500 microvolts; and 1, 2, 5, 10, 20 and 50 millivolts. It shall be possible to introduce a calibration signal into one or more channels without disturbing the recording of other channels.

3.4.5 Power supply cord. Power supply cord shall be 3-conductor 16 AWG or 18 AWG, Type SJ, or SJO with a 3 prong male plug, figure 4, style D of W-C-596.

3.5 Recorder.

3.5.1 Recording surface travel. The recording surface shall travel from the operator's right to his left and shall be visible for a distance of not less than 60 centimeters (cm) to the left of the recording line.

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3.5.2 Recording surface speed. The standard speed for recording shall be 30 millimeters (mm) per second plus or minus 2 percent. One low speed and one higher speed, each being an integral ratio of the standard, shall be provided (15 mm and 60 mm per second), capable of selection while the recorder is in motion.

3.5.3 Recording line. The width of the recorded line shall be not more than 1/2 mm in all directions of travel.

3.5.4 Spacing and registration of recording lines.

(a) At zero deflection, all the recorder writing points shall lie within ± 0.5 mm of a line perpendicular to the time axis of the record in the plane of the record.

(b) The spacing between writing points at zero deflection shall be not less than the maximum peak to peak recordable deflection.

(c) The maximum spacing of the recorders shall be not more than 35 mm. Each recorder shall be adjustable (to the accuracy specified) in respect to registration in line with the other recorders, lateral spacing between recorders, and pressure on the recording surface.

3.6 Accessories. The unit shall be supplied complete, ready to run with the following accessory items:

(a) Paper, 10 boxes, 5 packs of 1000 pages, folded
(b) Disc electrodes, 12 packages of 10 each
(c) Subdermal electrodes, 12 packages of 10 each
(d) Ear clip electrodes, 12 each
(e) Black recording ink, 6 bottles, 8 oz each
(f) Electrode paste, 6 tubes
(g) Frequency ruler, 2 each (per unit)
(h) Volt-ohmmeter

3.7 Finish. All painted metal parts shall be degreased and rustproofed prior to application of at least two coats of paint or enamel in the color normally furnished by the manufacturer.

3.8 Service data. The contractor shall enclose, with each unit, two copies each of one or more booklets which contain complete instructions for installation, operation, maintenance, and a list of component parts. Wiring diagrams shall be included for electrical items or components. Each part of the unit other than common fasteners, etc., shall be identified in the service data and parts list by the manufacturer's part number. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed service data, for review and comment by the Defense Medical Supply Center.

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The preliminary service data shall be submitted sufficiently early to permit adequate review by the Defense Medical Supply Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of required revisions and corrections in the final service data without delaying delivery under the contract or purchase order. In addition, the contractor shall mail 2 copies of the approved service data to the contracting officer for distribution.

3.9 Workmanship. Workmanship shall be first class throughout. The electroencephalograph shall be free from defects which detract from their appearance or may impair their serviceability.

3.10 Interchangeability of parts. All individual parts or assemblies of parts of any complete unit supplied on a contract shall be interchangeable with the corresponding parts or assemblies of parts of all other complete units supplied on the contract.

3.11 Identification marking. Each electroencephalograph shall bear the following information:

Federal stock number
Name or registered trademark of the manufacturer.
Model number.
Electrical data.
Contract or purchase order.
U.S.

Marking shall be legible and permanent and shall be accomplished by stamping or by the use of metal plates permanently affixed. Marking secured by adhesives shall not be used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling. Examination and testing shall be conducted on each unit to determine compliance with specification requirements.

4.3 Examination. The electroencephalograph shall be examined to determine compliance with all requirements in this specification.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. Level C. The subject commodity shall be packaged and packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.2 Marking.

5.2.1 Interior packages. Each package shall be marked to identify contents.

5.2.2 Exterior containers. Exterior containers shall be marked with stock number, name of material, quantity, name of consignee, name of consignor, and contract number as shown on contract or purchase order.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1).

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

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Military custodians:
Army - MD
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Review activities:
Army - MD
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Air Force - 03

Preparing activity:
Defense Supply Agency. - DM

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

FORM APPROVED BUDGET BUREAU NO. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use of the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION MILITARY SPECIFICATION MIL-E-36359(DM), dated 14 January 1965
ELECTROENCEPHALOGRAPH, 8 CHANNEL, 110 VOLT, 60 CYCLE, AC

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS (ATTACH ANY PERTINENT DATA WHICH MAY BE OF USE IN IMPROVING THIS SPECIFICATION. IF THERE ARE ADDITIONAL PAPERS, ATTACH TO FORM AND PLACE BOTH IN AN ENVELOPE ADDRESSED TO PREPARING ACTIVITY)

SUBMITTED BY (PRINTED OR TYPED NAME AND ACTIVITY)	DATE

DD FORM 1426

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